

Introduction to Boto3 Paginators and Why We need Paginators - Get All IAM Users using Client

- AWS serves millions of customers across the globe
- Suppose, If every customer were allowed to request all their data at once, AWS systems could become overloaded, leading to slowdowns or even outages
- To prevent this, when there is a possibility of returning a large amount of data in response to a customer request, AWS breaks the data into smaller chunks or "pages" – typically containing 100, 500, or 1000 results per page
- This is called pagination
- When we use a client method to execute read operations that involve large datasets then the client typically returns only the first page of results
- To retrieve the remaining data, we must explicitly read all pages by using special keys known as **pagination tokens** and **this process is called paginator**
- What is the use of Pagination ?
 - The use of pagination is to efficiently handle and retrieve large amounts of data from AWS services (or any API), without overloading the network or the system.
 - If AWS sent you thousands or even millions of records (like 10,000 EC2 instances or 1 million S3 objects) in a single API call then the response would be huge (potentially hundreds of MBs or even GBs of data)
 - It would take a lot longer to prepare the data and transmit it over the network
 - Large payloads are more prone to network failures (timeouts, dropped connections)